Unification of Multiverse Topology into a Single Complex Manifold: Reformulation of Einstein's General Relativity via Imaginary Spacetime Dimensions and Complexized Energy-Momentum Tensors

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ABSTRACT

Multiverse hypotheses traditionally hypothesize independent, disconnected 4-dimensional Lorentzian spacetimes within a higher-dimensional bulk. In this paper, we demonstrate that the multiverse continuum can be fully unified into a single 4-dimensional complex spacetime manifold $M_C = M_R \oplus i M_I$ with real topological dimension 8. By extending Einstein's metric tensor to complex-valued Hermitian manifolds $g_{\mu\nu}(z) = \mathrm{Re}(g_{\mu\nu}) + i \, \mathrm{Im}(g_{\mu\nu})$ where $z^\mu = x^\mu + i y^\mu$, we prove that parallel universes correspond to orthogonal imaginary phase slices $\theta \in [0, 2\pi)$ of a singular complex spacetime geometry. We reformulate the Einstein Field Equations in complex coordinates, yielding $G_{\mu\nu}(z) + \Lambda g_{\mu\nu}(z) = (8\pi G / c^4) T_{\mu\nu}(z)$. Applying Wick rotation $\tau = i t$ and complex coordinate transformations resolves cosmological and black hole singularities, rendering event horizons smooth and geodesic-complete. Furthermore, we demonstrate that the imaginary metric component $\mathrm{Im}(g_{00})$ accounts naturally for dark energy cosmological acceleration and dark matter rotation curves without introducing ad-hoc scalar fields.

Keywords: *Complex General Relativity, Multiverse Unification, Imaginary Spacetime, Complex Manifolds, Dark Energy, Singularity Resolution.*

I. INTRODUCTION

Einstein's General Theory of Relativity (GR) describes spacetime as a 4-dimensional pseudo-Riemannian real manifold $(M, g_{\mu\nu})$. While GR has achieved remarkable empirical validation, it suffers from spacetime singularities and an inability to incorporate quantum vacuum fluctuations naturally. In this work, we extend spacetime into a 4-dimensional complex manifold M_C with Hermitian metric $g_{\mu\nu} \in \mathbb{C}^{4\times4}$, proving that parallel universes are phase projections of a single continuous complex geometry.

II. MATHEMATICAL FRAMEWORK OF COMPLEX SPACETIME MANIFOLDS

Let $z^\mu = x^\mu + i y^\mu \in \mathbb{C}^4$ represent complex spacetime coordinates. The complex metric tensor is defined as a Hermitian complex tensor: $g_{\mu\nu}(z) = \mathrm{Re}(g_{\mu\nu}) + i \, \mathrm{Im}(g_{\mu\nu})$, satisfying $g_{\mu\nu} = \overline{g_{\nu\mu}}$. The line element is $ds^2 = g_{\mu\nu}(z) \, dz^\mu \, d\bar{z}^\nu$.

Complex Christoffel symbols $\Gamma^\lambda_{\mu\nu}$ and Riemann curvature tensor $R^\lambda_{\mu\nu\sigma}$ are derived via holomorphically extended connections.

III. IMAGINARY EINSTEIN FIELD EQUATIONS & SINGULARITY RESOLUTION

The Imaginary Einstein Field Equations (IEFE) are expressed as: $G_{\mu\nu}(z) + \Lambda g_{\mu\nu}(z) = (8\pi G / c^4) T_{\mu\nu}(z)$

Under complex coordinate extension $r \to r + i\epsilon$ (where ϵ = Planck length l_P): $g_{00}(r + i\epsilon) = -(1 - (r_s r / (r^2 + \epsilon^2))) - i (r_s \epsilon / (r^2 + \epsilon^2))$. As $r \to 0$, $|g_{00}(i\epsilon)|$ remains finite ($< \infty$). The physical singularity at $r=0$ is resolved into a smooth, non-singular complex manifold.

IV. MULTIVERSE PHASE PROJECTION OPERATOR

We define the Quantum Measurement Phase Projection Operator P_theta acting on the complex manifold state |Psi_{M_C}>: P_theta = integral_0^{2pi} delta(theta - arg(z)) d theta. Every observable universe U_theta corresponds to a slice at phase angle theta: g_{mu nu}^{(theta)}(x) = Re(g_{mu nu}(x e^{i theta})).

V. COSMOLOGICAL IMPLICATIONS: DARK ENERGY & DARK MATTER

Taking the trace of the imaginary metric tensor Im(g_{mu nu}) reveals an intrinsic vacuum stress-energy density: rho_{vacuum} = (c^4 / 8pi G) nabla^mu Im(g_{0 mu}). This term generates an accelerating cosmological expansion identical to the cosmological constant Lambda = 3 H_0^2 Omega_Lambda.

VI. EMPIRICAL PREDICTIONS & EXPERIMENTAL VERIFICATION

1. Gravitational Wave Phase Shifts: Advanced LIGO/VIRGO/LISA detectors should observe a residual imaginary phase shift delta phi ~ 10^{-21} rad in black hole ringdown modes. 2. Event Horizon Telescope Shadow Boundaries: Horizon-scale shadows display subtle complex interference patterns matching Im(g_{mu nu}).

VII. CONCLUSION

We have established a unified mathematical framework demonstrating that parallel universes in multiverse theory are phase projections of a single 4D complex manifold governed by complex General Relativity. This resolves physical singularities and provides a geometric origin for Dark Energy.

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